Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Plane Shapes: Their Types, Properties and How They Compare

A **plane shape** is a flat, two-dimensional (2-D) shape — it has length and width but no thickness. Every plane shape is made of straight sides (line segments) joined at their ends, and those joins are its corners, or **vertices**. The name of a shape usually tells you how many sides it has. The sides of a triangle mean "three angles", so a triangle has **3 sides and 3 vertices**.

Here are the common plane shapes and their key properties:

- **Triangle** — 3 straight sides and 3 vertices. It has 3 angles.
- **Quadrilateral** — any shape with **4 straight sides and 4 vertices**. Square, rectangle, parallelogram and trapezium are all quadrilaterals.
- **Square** — a quadrilateral with all 4 sides **equal** and all 4 angles **right angles** (90°).
- **Rectangle** — a quadrilateral with 4 right angles, but only the **opposite** sides equal.
- **Parallelogram** — a quadrilateral with **opposite sides equal and parallel**. Opposite angles are equal.
- **Trapezium** — a quadrilateral with **one pair of opposite sides parallel**.
- **Pentagon** — 5 straight sides and 5 vertices.
- **Hexagon** — 6 straight sides and 6 vertices.
- **Octagon** — 8 straight sides and 8 vertices.
- **Circle** — the one plane shape without straight sides. It is a curved closed line, and every point on it is the same distance from the centre.

You can see the **similarities and differences** between shapes by comparing these properties. For example, a square and a rectangle are alike because both have 4 sides and 4 right angles. They differ because a square has all 4 sides equal, while a rectangle only has opposite sides equal. So every square is a rectangle, but not every rectangle is a square.

Look at how the shapes are related to the number of their sides:

- 3 sides — triangle
- 4 sides — quadrilateral (square, rectangle, parallelogram, trapezium)
- 5 sides — pentagon
- 6 sides — hexagon
- 8 sides — octagon

A shape whose sides are all equal and whose angles are all equal is called a **regular** shape. A square is a regular quadrilateral because all its sides and angles are equal; a rectangle is **not** regular because its four sides are not all equal.

Worked example 1 — naming a shape by its properties. A shape has 4 straight sides, 4 right angles, and all 4 sides are equal. What is it?

Count the sides: it has 4 sides, so it is a quadrilateral. Among the quadrilaterals, the one with 4 right angles **and** all 4 sides equal is the **square**.

Worked example 2 — comparing a parallelogram and a trapezium. How are they alike, and how do they differ?

Both are quadrilaterals, so both have 4 sides and 4 vertices. They differ in which sides are parallel: a parallelogram has **two** pairs of opposite sides parallel, while a trapezium has only **one** pair of opposite sides parallel.

Worked example 3 — the number of sides tells the name. A shape has 8 straight sides. What is it called?

The shape with 8 sides is an **octagon**. It has 8 vertices and 8 angles.

Practice

1. State the number of sides and vertices of each of these plane shapes: (a) a hexagon, (b) a pentagon, (c) a triangle.
2. A shape has 4 straight sides and 4 right angles, but its four sides are not all equal. What is the shape called, and why is it not a square?
3. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all the non-adjacent vertices?

Section 2: The Parts of a Circle

A circle is a flat, closed shape made up of every point that is the same distance from one fixed point inside it, called the centre. Because a circle has no straight sides and no corners, it has its own special parts with their own names. Learning these names lets you describe a circle exactly and talk about how it is built.

Here are the parts you need to know.

- **Centre**: the fixed point in the very middle of the circle. Every point on the circle is the same distance from it.
- **Circumference**: the distance all the way around the circle. It is the circle's own "perimeter" — its curved outer boundary.
- **Radius** (plural *radii*): a straight line from the centre to any point on the circumference. All the radii of one circle are equal in length.
- **Diameter**: a straight line that passes through the centre and touches the circumference at both ends. It is twice as long as the radius.
- **Chord**: a straight line joining any two points on the circumference. A chord does not have to pass through the centre.
- **Arc**: part of the circumference — a curved piece of the circle's boundary.
- **Segment**: the region inside the circle bounded by a chord and the arc it cuts off. (Picture the piece "cut off" from the rest of the circle.)
- **Sector**: the region inside the circle bounded by two radii and the arc between their ends. It looks like a slice or a wedge of the circle.
- **Tangent**: a straight line that touches the circumference at exactly one point and does not cross into the circle.

Let us name the parts on a circle you can picture clearly.

Think of a circle with a point C in its very middle. From C, draw a straight line out to a point A on the circle's edge. This line CA is a **radius**, because it runs from the centre to the circumference.

Now draw another straight line from a point B on the circle, straight through the centre C, to a point D on the other side. The whole line BD passes through the centre and meets the circle at both ends, so BD is a **diameter**. Notice BD is made of two radii placed end to end, which is why a diameter is twice a radius.

The distance all the way around the circle from A, round and back to A, is the **circumference**. Take just the curved piece of the edge from A to B — that curved piece is an **arc**.

Join A and B with a straight line inside the circle. That line is a **chord** (here it does not pass through the centre, so it is not a diameter). The chord AB together with the arc AB closes off a region inside the circle — that region is a **segment**.

Now draw a straight line from the centre C to A, and another from C to B. These two radii, CA and CB, together with the arc AB between their ends, enclose a wedge-shaped region — that region is a **sector**.

Finally, imagine a straight line that just touches the circle at point A, resting against the edge without entering it. That line is a **tangent** to the circle at A.

So in one circle you have: the centre C, a radius CA, a diameter BD, the circumference around the edge, an arc AB, a chord AB, a segment between chord AB and arc AB, a sector between radii CA and CB and arc AB, and a tangent touching at A.

Practice

1. Which part of a circle is the distance all the way around it?

2. A straight line from the centre of a circle to a point on the circumference is called the _____.
3. How does a diameter compare in length with a radius of the same circle?
4. A straight line joining any two points on the circumference of a circle is called a _____. When must this line also be a diameter?
5. Match each part of a circle with its meaning: arc, segment, sector, tangent.
   (a) A straight line that touches the circumference at exactly one point.
   (b) A curved piece of the circumference.
   (c) A region bounded by two radii and the arc between them.
   (d) A region bounded by a chord and the arc it cuts off.
6. In one circle, the radius is 5 cm. What is the length of the diameter? Explain your answer.

Section 3: Finding the Perimeter of Plane Shapes

The perimeter of a plane shape is the total distance all the way around its outside edge. To find it you simply add up the lengths of every side.

Different shapes have different formulas, because their sides are related in different ways.

Square — all four sides are equal, so P = 4 × side.
Rectangle — the two lengths are equal and the two breadths are equal, so P = 2 × (length + breadth).
Parallelogram — opposite sides are equal, so P = 2 × (base + side).
Trapezium — all four sides may be different, so just add all four: P = a + b + c + d.
Regular polygon — all sides are equal, so P = (number of sides) × (length of one side).
Circle — its perimeter is called the circumference, and C = 2 × π × radius (often written C = 2πr).

For a circle you will usually take π = 22/7. Remember that the radius is the distance from the centre to the edge, and the diameter is twice the radius.

Worked Example 1 — adding all the sides
A parallelogram has a base of 6 cm and a side of 5 cm. Find its perimeter.

A parallelogram has opposite sides equal, so the two bases are both 6 cm and the two sides are both 5 cm.
Perimeter = 2 × (base + side) = 2 × (6 + 5) = 2 × 11 = 22 cm.

Worked Example 2 — a regular polygon
A regular hexagon has six equal sides, each 5 cm. Find its perimeter.

Perimeter = number of sides × length of one side = 6 × 5 = 30 cm.

Worked Example 3 — a circle and finding the radius
The circumference of a circle is 22 cm. Find its radius. Take π = 22/7.

C = 2πr, so 22 = 2 × (22/7) × r.
Divide both sides by 2: 11 = (22/7) × r.
Multiply both sides by 7/22: r = 11 × 7/22 = 77/22 = 3.5.
So the radius is 3.5 cm.

Worked Example 4 — finding a missing side when the perimeter is known
A rectangle has perimeter 24 cm and its breadth is 4 cm. Find its length.

P = 2 × (length + breadth), so 24 = 2 × (length + 4).
Divide both sides by 2: 12 = length + 4.
Subtract 4: length = 8 cm.

Practice

1. Find the perimeter of a square with sides 8 cm.

2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
3. What is the perimeter of a regular pentagon with sides 5 cm?
4. Find the perimeter of a parallelogram whose base is 6 cm and sides are 5 cm.
5. The circumference of a circle is 22 cm. Find the radius of the circle. (Take $\pi = 22/7$.)
6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle (its perimeter).
7. Find the perimeter of a regular hexagon with sides 5 cm.

Check your work

Attempt every question before you read this page.

Plane Shapes: Their Types, Properties and How They Compare

1. (a) A hexagon has 6 sides and 6 vertices. (b) A pentagon has 5 sides and 5 vertices. (c) A triangle has 3 sides and 3 vertices.

   The number of vertices is always the same as the number of sides. Hexagon — "hex" means six, so 6 sides and 6 vertices; pentagon — "pent" means five, so 5 sides and 5 vertices; triangle — "tri" means three, so 3 sides and 3 vertices.
2. It is a rectangle. It is not a square because a square must have all 4 sides equal, and this shape does not.

   Both a square and a rectangle have 4 right angles. The difference is in the sides: a square has all 4 sides equal, while a rectangle has only opposite sides equal. Since this shape's sides are not all equal, it cannot be a square, so it must be a rectangle.
3. 6 triangles

   From one vertex of a shape with n sides, you can draw diagonals to every vertex except itself and its two neighbours — that is $n - 3$ diagonals, which split the shape into $n - 2$ triangles. An octagon has $n = 8$, so it splits into $8 - 2 = 6$ triangles.

The Parts of a Circle

1. The circumference.
2. Radius.
3. The diameter is twice as long as the radius.
4. A chord; it is also a diameter when it passes through the centre.
5. (a) tangent, (b) arc, (c) sector, (d) segment.
6. 10 cm, because the diameter is twice the radius (2×5 cm $= 10$ cm).

   The diameter passes through the centre and touches the circumference at both ends, so it is made of two radii placed end to end. Radius = 5 cm, so diameter $= 2 \times 5$ cm $= 10$ cm.

Finding the Perimeter of Plane Shapes

1. 32 cm

   $P = 4 \times \text{side} = 4 \times 8 = 32$ cm.
2. 22 cm

   $P = 2 \times (\text{length} + \text{breadth}) = 2 \times (7 + 4) = 2 \times 11 = 22$ cm.
3. 25 cm

   A pentagon has 5 equal sides. $P = 5 \times 5 = 25$ cm.
4. 22 cm

   $P = 2 \times (\text{base} + \text{side}) = 2 \times (6 + 5) = 2 \times 11 = 22$ cm.
5. 3.5 cm

   $C = 2\pi r$, so $22 = 2 \times (22/7) \times r$. Divide both sides by 2: $11 = (22/7) \times r$. Multiply by 7/22: $r = 11 \times 7/22 = 77/22 = 3.5$ cm.
6. 6d

   Breadth = d, so length = 2d. Perimeter $= 2 \times (\text{length} + \text{breadth}) = 2 \times (2d + d) = 2 \times 3d = 6d$.

7. 30 cm

   A hexagon has 6 equal sides. P = 6 × 5 = 30 cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-53601b3e	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____